CANONICAL ISOMORPHISM BETWEEN L-ETA_I(M) TENSORED WITH O_X/I AND THE BOCKSTEIN COMPLEX H^BULLET(M/I)

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites four cross-chapter labels (including a situation and an equation), each recorded in assembly.external_references as a navigation pointer with its canonical tag.

Editorial note. The proof cites no bibliography item inside the counted span.

Allowed standard prerequisites (editorial).

- The functor L-eta_I and its computation on torsion free complexes. Stacks Project, Cohomology of Sheaves, Lemma lemma-eta-qis (tag 0F8P) and Lemma lemma-Leta of this chapter. Justifies replacing M by a complex of I-torsion free modules and computing L-eta_I there.
- Algebraic eta second property. Stacks Project, More on Algebra, Lemma more-algebra-lemma-eta-second-property (tag 0F7T). The stalkwise model that the constructed map of complexes is compared against.

Source reference. Original reference 0GT4: sources/stacks-chow/tags.txt, source lines 20954–20954. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original equation 0GT8: sources/stacks-chow/tags.txt, source lines 20958–20958. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0F8P: sources/stacks-chow/tags.txt, source lines 19049–19049. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0F7T: sources/stacks-chow/tags.txt, source lines 19018–19018. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *In Situation 0GT4 let M be an object of $D(\mathcal{O}_X)$. There is a canonical isomorphism*

$$L\eta_{\mathcal{I}}M \otimes^{\mathbf{L}} \mathcal{O}_X/\mathcal{I} \longrightarrow H^\bullet(M/\mathcal{I})$$

in $D(\mathcal{O}_X)$ where the right hand side is the complex (0GT8).

Proof. By the construction of $L\eta_{\mathcal{I}}$ in Lemma 0F8P we may assume M is represented by a complex of $\mathcal{I}$-torsion free $\mathcal{O}_X$-modules $\mathcal{F}^\bullet$. Then $L\eta_{\mathcal{I}}M$ is represented by the complex $\eta_{\mathcal{I}}\mathcal{F}^\bullet$ which is a complex of $\mathcal{I}$-torsion free $\mathcal{O}_X$-modules as well. Thus $L\eta_{\mathcal{I}}M \otimes^{\mathbf{L}} \mathcal{O}_X/\mathcal{I}$ is represented by the complex $\eta_{\mathcal{I}}\mathcal{F}^\bullet \otimes \mathcal{O}_X/\mathcal{I}$. Similarly, the complex $H^\bullet(M/\mathcal{I})$ has terms $H^i(\mathcal{F}^\bullet \otimes \mathcal{I}^i/\mathcal{I}^{i+1})$.

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Let f be a local generator for $\mathcal{I}$. Let s be a local section of $(\eta_{\mathcal{I}}\mathcal{F})^i$. Then we can write $s = f^i s'$ for a local section s' of $\mathcal{F}^i$ and similarly $d^i(s) = f^{i+1}t$ for a local section t of $\mathcal{F}^{i+1}$. Thus d^i maps $f^i s'$ to zero in $\mathcal{F}^{i+1} \otimes \mathcal{I}^i/\mathcal{I}^{i+1}$. Hence we may map s to the class of $f^i s'$ in $H^i(\mathcal{F}^\bullet \otimes \mathcal{I}^i/\mathcal{I}^{i+1})$. This rule defines a map

$$(\eta_{\mathcal{I}}\mathcal{F})^i \otimes \mathcal{O}_X/\mathcal{I} \longrightarrow H^i(\mathcal{F}^\bullet \otimes \mathcal{I}^i/\mathcal{I}^{i+1})$$

of $\mathcal{O}_X$-modules. A calculation shows that these maps are compatible with differentials (essentially because β sends the class of $f^i s'$ to the class of $f^{i+1}t$), whence a map of complexes representing the arrow in the statement of the lemma.

To finish the proof, we observe that the construction given in the previous paragraph agrees on stalks with the maps constructed in More on Algebra, Lemma 0F7T hence we conclude. $\square$